Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/20/2015
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening in Solid-Liquid Mixtures (CSLM)-2: NNX07AW01G		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2007	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 9/30/2014 per NSSC information (Ed., 10/25/2012) NOTE: End date is now 9/30/2012 per NSSC information (Ed., 6/7/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AW01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Previous NASA sponsored work clearly showed that solid-liquid mixtures consisting of Sn-rich particles in a Pb-Sn eutectic liquid are ideal, and perhaps unique, systems in which to explore the dynamics of the Ostwald ripening process. The high coarsening rate in these systems permit accurate kinetic data to be obtained, and the thermophysical parameters necessary to make a comparison between theory and experiment are known. However, in a terrestrial environment experiments can be performed only at the relatively high volume fractions of solid where the presence of a solid skeletal structure prevents large-scale particle sedimentation. In these high volume fraction experiments, a comparison between theory and experiment shows that the solid-liquid mixtures are coarsening faster than predicted by an approximate theory for purely diffusional controlled Ostwald ripening. The objective of this project is to plan and perform a microgravity experiment on Ostwald ripening in solid-liquid mixtures. This experiment will serve two primary purposes: it will allow experiments to be performed which can be directly compared to heretofore untested theories for coarsening in systems with low volume fractions of solid, and it will eliminate conclusively convection of the liquid matrix and small-scale particle motion within the skeletal structure as possible sources for the disagreement observed between theory and experiment in the high volume fraction experiments. To perform coarsening experiments whose results can be compared directly with theory with no adjustable parameters. The spaceflight experiment will produce data which, for the first time, can be compared directly to theory with no adjustable parameters. This data will address the long standing controversy over the dependence of the coarsening rate of a two-phase system on the volume fraction of coarsening phase.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Coarsening in Solid-Liquid Mixtures (CSLM)-2 will make possible predictions of coarsening kinetics in alloys with high volume fractions of precipitates, such as alloys under consideration for high temperature exploration applications (i.e., nuclear propulsion, waste heat coolant loops). Results will be applied to materials design codes that will reduce cost, shorten development time, and improve performance over conventional materials design methods.
Task Progress:	We have completed our analysis of the lower volume fraction samples, in particular the 15, 20, and 30% volume fraction samples. The predictions of the theory have been confirmed in the 15 and 20% samples. The results from the 30% volume fraction samples show that there is a major difference between the predictions of theory and experiment. A simulation of the effects of residual accelerations on the shuttle on the coarsening process has been developed as well. The simulations show that the very slow motion of the particles induces changes in the rate constant and particle size distributions in a manner consistent with experiment. Finally, experiments on the effects of residual accelerations on the dynamics of particle motion show that during experiments on the International Space Station (ISS) the two-phase mixtures were coarsening in a diffusion-limited manner.
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Genau AL, Voorhees PW, Thornton K. "The morphology of topologically complex interfaces." Scripta Materilia. 2009 Mar;60(5):301-4. http://dx.doi.org/10.1016/j.scriptamat.2008.10.032 , Mar-2009
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Articles in Peer-reviewed Journals	Kammer D, Voorhees PW. "Analysis of complex microstructures: Serial sectioning and phase-field simulations." MRS Bulletin. 2008 Jun;33(6):603-10. http://dx.doi.org/10.1557/mrs2008.125 , Jun-2008
Articles in Peer-reviewed Journals	Thompson JD, Gulsoy EB, Voorhees PW. "Self-similar coarsening: A test of theory." Acta Materialia. 2015 Nov;100:282-9. https://doi.org/10.1016/j.actamat.2015.08.036 , Nov-2015
Papers from Meeting Proceedings	Kammer D, Genau A, Voorhees PW, Duval WM, Hawersaat RW, Lorik JM, Hall DG, Frey CA. "Results from the International Space Station: Coarsening in Solid-Liquid Mixtures." 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, January 5-9, 2009, Orlando, FL. 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, January 5-9, 2009, Orlando, FL. AIAA paper 2009-616. http://dx.doi.org/10.2514/6.2009-616 ; accessed 5/20/15. , Jan-2009
Papers from Meeting Proceedings	Duval WMB, Hawersaat RW, Lorik T, Thompson J, Gulsoy B, Voorhees PW. "Coarsening in solid-liquid mixtures: overview of experiments on shuttle and ISS." MS&T 2013-Materials Science & Technology 2013, Montreal, Quebec, Canada, October 27-31, 2013. In: Proceedings Materials Science & Technology Conference and Exhibition 2013 (MS&T'13). Warrendale, PA: Association for Iron & Steel Technology, 2013. Volume 4, p. 2745-2752. Printed by Curran Associates, Inc. 2013. , Oct-2013